

Reinforcement Learning

End Semester Exam

05/12/2023

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Instructions: You have two hours to work on the questions. Answers without supporting steps will receive zero credit. Any resources, other than a pen/pencil, are **not** allowed. In case you believe that required information is unavailable, make a suitable assumption.

Question 1. 10 marks Consider range space $S_X = \{0, 1, 2\}$ of random variable X . Derive the probability distribution, specifically the probability mass function, $p(x, \theta)$, parametrized by an appropriate choice of the vector θ . [Hint: You may want to use the Kronecker delta function $\delta_{ij} = 1, i = j$, and $\delta_{ij} = 0$, otherwise.]

Illustrate the region in which points correspond to the probability distributions $p(x, \theta)$.

Question 2. 10 marks Consider a probability distribution $p(x, \theta)$ parametrized by the vector θ . Show that

$$E \left[\frac{\partial \log(p(x, \theta))}{\partial \theta_i} \frac{\partial \log(p(x, \theta))}{\partial \theta_j} \right] = -E \left[\frac{\partial}{\partial \theta_i} \frac{\partial \log(p(x, \theta))}{\partial \theta_j} \right]. \quad (1)$$

Question 3. 5 marks Simplify the expected value

$$E \left[\frac{\partial \log(p(x, \theta))}{\partial \theta_i} \right]. \quad (2)$$

Question 4. 20 marks Consider the curve $\theta(t)$. Points on the curve correspond to points on a statistical manifold. Let the statistical manifold be defined by the set of probability distributions $p(x, \theta)$ parametrized by θ . We are interested in optimizing a certain function $f(\theta)$. Assume that θ is $n \times 1$. Answer the following questions.

- (5 marks) Write down the directional derivative of $f(\theta)$ along vector $\vec{v}$.
- (5 marks) Consider the basis $\left\{ \frac{\partial \log(p(x, \theta))}{\partial \theta_1}, \dots, \frac{\partial \log(p(x, \theta))}{\partial \theta_n} \right\}$. Derive the squared norm of $\vec{v}$ using the representation of $\vec{v}$ in terms of the above basis and the definition of the inner product.
- (10 marks) We want to find the vector $\vec{v}$ that minimizes the directional derivative, under the constraint that the norm of $\vec{v}$ is 1. Let Cf denote the directional derivative that you must have calculated above. The necessary first order conditions are obtained by setting the partial derivatives of $Cf - \lambda|\vec{v}|^2$, with respect to the elements of $\vec{v}$, to zero. Derive the n equations that constitute the first order conditions. Write the equations in terms of the Fischer Information matrix G , the $(i, j)^{\text{th}}$ element of which is given by Equation 1. Use the equations to calculate the vector $\vec{v}$ that minimizes the directional derivative.

Question 5. 15 marks Write down the expected squared approximation error in the action-value function, when using policy π . Define any other needed distributions to write the expectation. Assume your approximation is parametrized by the vector θ . Derive the first order conditions that are satisfied where the error is minimized (at a local minimum). Calculate the minimizing θ when the approximation is linear with feature matrix A . Specify the dimensionality of A and θ .

Question 6. 15 marks We want to minimize

$$E_{X \sim P_X(x)} [L(X, \theta)], \quad (3)$$

with respect to the parameter vector θ , in an online data-driven manner. $L(X, \theta)$ is a certain loss function.

You are tasked with designing an algorithm that updates θ every time a new data sample is obtained, resulting in n updates starting with an initial vector θ_0 .

An added challenge that your algorithm must address is that samples received by the algorithm are drawn from the PMF $P'_X(x) = P[X' = x]$ of random variable X' instead of $X \sim P_X(x)$. Assume that the algorithm knows both the PMFs $P'_X(x)$ and $P_X(x)$. Further assume that the algorithm receives the n data samples $x_1, x_2, \dots, x_n$, respectively, at the beginning of iterations $1, 2, \dots, n$.

Provide the n iterations of your algorithm. Explain the workings of your algorithm and why it solves the given minimization problem.

Question 7. 10 marks Expand $E[G_t | S_{t+1} = s']$ in terms of the reward received at $t + 1$, the MDP functions, the value function, the policy π , and any other PMFs that may be required.

Question 8. 15 marks Consider a MDP with two states 0 and 1. An agent may choose to stay in the current state or move to the other state. The reward of staying in the same state is 2 and that of moving to the other state is 1. Show two policy iteration steps, while restricting your choice of policies to the set of ϵ -soft policies. Assume $\epsilon = 0.1$ and $\gamma = 1$.